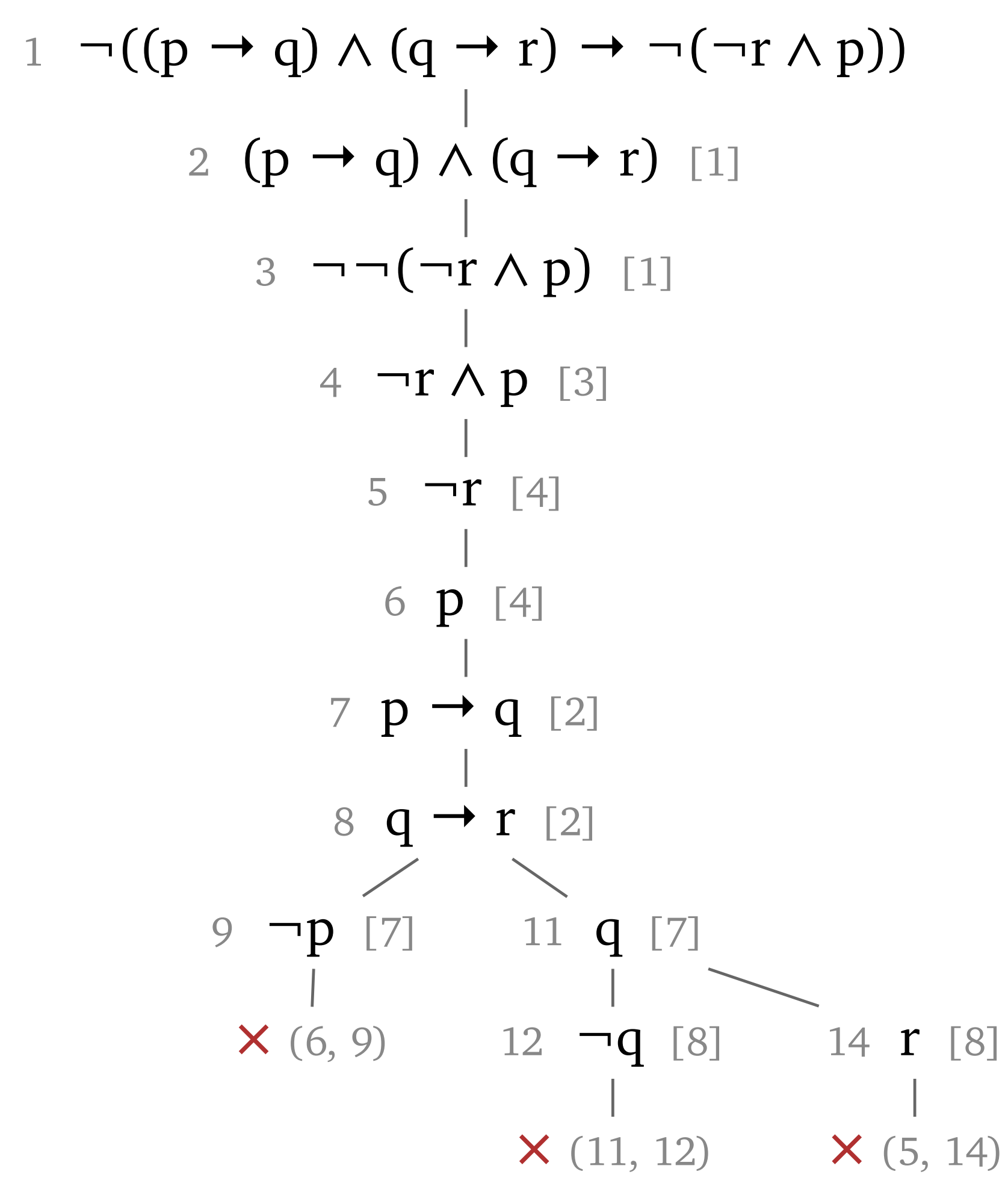
1 ¬((p → q) ∧ (q → r) → ¬(¬r ∧ p))
2 (p → q) ∧ (q → r) [1]
3 ¬¬(¬r ∧ p) [1]
4 ¬r ∧ p [3]
5 ¬r [4]
6 p [4]
7 p → q [2]
8 q → r [2]
9 ¬p [7]
11 q [7]
× (6, 9)
12 ¬q [8]
14 r [8]
× (11, 12)
× (5, 14)